

CORE ENGINE & ARCHITECTURE BIBLE

PRANGARA - BACKEND

HANDBOOK

Deep-Dive Engineering Guide: From Raw Bill Pixels & Metal Nameplates to Deterministic GHG Math, MACC Economics, and Global Export Compliance

⚙️ FastAPI
Python 3.11+

📐 Zero-LLM
Deterministic Math

👁️ Local Ollama
VLM Ingestion

📊 MACC
Optimization
Engine

🛡️ Pydantic
Strict Invariants

An exhaustive technical specification written for engineers, technical auditors, climate scientists, and curious beginners.

PRANGARA BACKEND ARCHITECTURE SPECIFICATION · REPO: [/backend/app](#) & [/backend/engine](#)

1. The Processing Pipeline (Toddler to Engineer)

🧸 Toddler Version:

Imagine you give a messy toy box to a robot. First, a camera robot looks at the box and writes down how many blocks are red and blue. Next, a strict teacher checks if any blocks are missing or broken. Then, a giant, super-fast calculator multiplies the blocks by secret government numbers. Finally, a smart doctor says: "Aha! You have too many broken blocks here—let's fix these three, and you will get ₹50 Lakhs back!"

```
[STEP 1: RAW INGESTION] Electricity Bills (PDF/JPG) → Ollama VLM (Local OCR) Motor Nameplate Photos → Vision  
Extractor Fuel Delivery Challans → Heuristic Regex Parsers Excel / CSV Records → Columnar Schema Mapper | ▼  
[STEP 2: THE CANONICAL ADAPTER & PYDANTIC BOUNCER] Validates into: PlantProfileIn (Strict type check, NO negative  
values, non-null guarantees) | ▼ [STEP 3: THE DETERMINISTIC FACTOR REGISTRY] Matches streams with FactorDB: - CEA  
v22 Regional Grid Factor (0.716 kg CO2e/kWh) - IPCC 2006 Fuel Combustion Tables & NCV Energy Densities (GJ) - DEFRA  
Modal Freight Emission Coefficients (tCO2e / t.km) | ▼ [STEP 4: STREAM-RESOLVED FOOTPRINT ENGINE] - Scope 1:  
Aggregated Process Heat (Coal, Briquette, Gas) + Standalone Diesel - Scope 2: Purchased Grid Electricity - Scope 3:  
Purchased Materials + Waste + Inbound/Outbound Freight - Biogenic CO2 memo accounting (GHG Protocol compliance) | ▼  
[STEP 5: BENCHMARK-RELATIVE LEAK DETECTOR] Compares Gate-to-Gate (Scope 1+2) intensity vs. BEE 55 MSME Sector  
Benchmarks: - Rule 1: Benchmark Breach (> p75 of sector) → Critical/High severity - Rule 2: Material  
Concentration (> 15% of total & > p50) - Rule 3: Structural Hotspot (> 25% of total, purchased Scope 3 materials) |  
▼ [STEP 6: MARGINAL ABATEMENT COST CURVE (MACC)] For each candidate intervention: - Annualises Capex with Capital  
Recovery Factor:  $CRF = r(1+r)^n / ((1+r)^n - 1)$  - Computes Levelized Cost of Abatement:  $LCOA = (CRF \cdot Capex + \Delta Opex - Saving) / Abatement$  - Ranks into Cash-Positive ( $LCOA < 0$ ), Quick-Wins, and Capital Projects | ▼ [STEP 7:  
COMPLIANCE, SIMULATION & DISPATCH] - EU CBAM Export Exposure:  $(Scope\ 1 + Scope\ 2) \times EU\ Export\ Share \times \text{€ Reference Price}$  - Green Logistics: Corridor consolidation, Cargo Pooling, Modal Shift (Road to Rail/EV) - Industrial  
Symbiosis: By-product matchmaking with Avoided Embodied Emissions - Server-Side Vector SVG Report Generator (1-  
Click Auditable PDF/CSV Export)
```

2. Methods of Extraction: How Messy Papers Turn into Data

1. Local Ollama VLM (Vision AI)

ON-PREMISE

Located in `backend/app/services/ollama_service.py`.
Interacts with local vision models (`llama3` , `gemma` , `llava`)
running at `http://127.0.0.1:11434` .

- **Utility Bills:** Scans electricity bills (TANGEDCO, MSEDCL, BESCOM) extracting: *Billing Period, Total kWh, Billed Demand (kVA), Power Factor, and Net Payable*.
- **Motor Plates:** Reads stamped metal: *Rated kW, RPM, Efficiency Class (IE1/IE2/IE3/IE4), Voltage*.
- **100% Data Sovereignty:** Zero cloud API calls. Confidential plant figures never leave the factory computer.

2. Heuristic Pattern Matchers (Regex)

ZERO-LLM

Located in `backend/app/services/intake_extract.py`.
Deterministic fallback that operates with 100% reproducibility.

- **Regex Token Extraction:** Identifies meter serials, fuel liters, coal metrics (MT), and billing dates.
- **Confidence Scoring:** Computes a data-quality confidence index (0.0 to 1.0) based on document completeness.
- **Sanity Clamps:** Flags suspicious entries (e.g. power factor > 1.0 or negative kilowatt-hours).

The Strict Gatekeeper: The `PlantProfileIn` Canonical Contract

In `backend/app/schemas/factory.py` , every intake path (form, OCR, mobile app, CSV) must serialize into the exact dataclass consumed by the calculation engine. If an AI invents a non-existent field, Pydantic immediately aborts the pipeline:

```
class PlantProfileIn(ApiModel):
    annual_output_t: float = Field(default=0, ge=0)
    electricity_kwh: float = Field(default=0, ge=0)
    fuels: dict[str, float] = Field(default_factory=dict)
    materials: dict[str, float] = Field(default_factory=dict)
    waste: dict[str, float] = Field(default_factory=dict)
    freight: dict[str, float] = Field(default_factory=dict)

    @model_validator(mode="after")
    def _no_negative_quantities(self) -> "PlantProfileIn":
        for label, mapping in (("fuels", self.fuels), ("materials", self.materials), ...):
            for key, val in mapping.items():
                if val < 0: raise ValueError(f"{label}.{key} cannot be negative")
```

3. The Exact Mathematical Calculations (No Black Box)

A. Scope 2: Purchased Electricity Calculation

Purchased grid electricity emissions are calculated using state and regional CEA baseline factors:

```
$$E_{\text{Scope2}} = \frac{\text{Electricity (kWh)}}{\text{Grid Factor}} \times (\text{tCO}_2\text{e/MWh}) \times 1000 \quad \text{Example:} \\ 500,000 \text{ kWh} \times 0.716 \text{ kg/kWh} / 1000 = \mathbf{358.00 \text{ tCO}_2\text{e}}$$
```

B. Scope 1: Thermal Fuel Aggregation & NCV Energy Conversion

Different fossil fuels (Indian coal, biomass briquettes, furnace oil, natural gas) have different energy densities. Prangara normalizes them all into ****Gigajoules (GJ)**** using Net Calorific Values (NCV):

Fuel Key	Net Calorific Value (NCV)	IPCC Emission Factor	Standard Unit
COAL_INDIAN	15.9 GJ / tonne	94.6 kg CO ₂ e / GJ	Metric Tonne
BIOMASS_BRIQUETTE	15.0 GJ / tonne	Exempt from Scope 1 total (Biogenic)	Metric Tonne
FURNACE_OIL	0.0405 GJ / kg	77.4 kg CO ₂ e / GJ	Kilograms
NATURAL_GAS	0.0360 GJ / Sm ³	56.1 kg CO ₂ e / GJ	Standard m ³
DIESEL	0.0360 GJ / Litre	74.1 kg CO ₂ e / GJ (2.68 kg CO ₂ /L)	Litre

$$\text{Thermal Energy (GJ)} = \sum \left(\text{Fuel Qty}_i \times \text{NCV}_i \right)$$

$$\text{Process Heat Emissions } (\text{tCO}_2\text{e}) = \sum \left(\text{Fuel Qty}_i \times \text{Factor}_i \right)$$
Critical GHG Protocol Rule: Biomass briquette direct CO₂ emissions are **biogenic**. Under GHG Protocol, they are excluded from the Scope 1 total and reported on a transparent separate memo disclosure line: $\text{Biogenic } \text{CO}_2 = \text{Qty} \times 1.55 \times \text{tCO}_2 / \text{t}$.

4. How Prangara "Predicts" & Diagnoses (The Heuristic Logic)

The Gate-to-Gate vs. Cradle-to-Gate Invariant

A fundamental mistake in amateur carbon calculators is comparing total factory emissions against published energy benchmarks. Prangara enforces a strict separation:

Gate-to-Gate Intensity (Scope 1 + Scope 2):

$$\frac{\text{Scope 1} + \text{Scope 2}}{\text{Annual Production (Tonnes)}}$$

This represents what the plant operations actually control inside their four walls (boilers, furnaces, motors). **This is the ONLY part benchmarked against sector peers.**

Cradle-to-Gate Footprint (Scope 1 + 2 + 3):

$$\frac{\text{Scope 1} + \text{Scope 2} + \text{Scope 3}}{\text{Annual Production (Tonnes)}}$$

Includes purchased raw materials, packaging, and freight. A plant that purchases more steel per unit is not running a worse machine shop; mixing the two produces false penalties.

The 3 Benchmark-Relative Leak Detection Rules

In `backend/engine/leaks.py`, a "leak point" is not a physical gas leak—it is a point where financial capital and carbon escape unnecessarily relative to industry peers:

Rule Name	Trigger Condition	Severity Assigned	Real-World Industrial Example
1. Benchmark Breach	Plant's stream intensity exceeds the 75th percentile (p75) of sector peers.	CRITICAL / HIGH	A textile dyeing unit consumes 780 kWh/t when the Surat cluster p75 is 620 kWh/t (indicating low-efficiency motors or pump throttling).
2. Material Concentration	Stream represents > 15% of total footprint and sits above the 50th percentile (p50 median) .	MODERATE	A foundry's cupola furnace coal consumption is 28% of total emissions and slightly above peer median (high-impact optimization candidate).
3. Structural Hotspot	Stream exceeds > 25% of total footprint with no benchmark available (Scope 3 materials).	WATCH	An auto-parts stamper whose virgin CRCA steel sheets represent 72% of total emissions (action: substitute secondary scrap steel).

5. Marginal Abatement Cost Economics (MACC Engine)

Located in `backend/engine/macc.py`. How does the backend rank what a factory should do first? It calculates the **Levelized Cost of Abatement (LCOA)**:

$$LCOA = \frac{\left(CRF \times \text{Capex} \right) + \Delta \text{Opex} - \text{Gross Annual Saving}}{\text{Annual Abatement} \times (t_{CO_2e})}$$

1. Capital Recovery Factor (CRF): The Honest Way to Annualize Capex

Amateur calculators charge 100% of Capex to Year 1, making a 25-year solar plant look financially terrible compared to cheap LED bulbs. Prangara annualizes capital using the discount rate r (default 12% MSME cost of capital) and asset life n :

$$CRF = \frac{r(1+r)^n}{(1+r)^n - 1}$$

Example: At 12% discount rate over 15-year life: $CRF = \frac{0.12(1.12)^{15}}{(1.12)^{15} - 1} = 0.1468$

A ₹50 Lakh solar plant has an annualized capital charge of: $50,00,000 \times 0.1468 = ₹7,34,000 / \text{year}$.

2. Net Annual Benefit & Simple Payback

$$\begin{aligned} \text{Net Annual Benefit (\text{\text{₹}}/\text{yr})} &= \text{Gross Energy Saving} - \left(\text{CRF} \times \text{Capex} \right) - \Delta \text{Opex} \\ \text{Simple Payback (Years)} &= \frac{\text{Capex}}{\text{Gross Annual Saving} - \Delta \text{Opex}} \end{aligned}$$

3. The Golden MACC Curve Ranking Rule

- **Negative LCOA (\$ < 0\$):** The intervention generates more annual cash savings than its annualized cost. **It puts money in the owner's bank account every single year.**
- **Positive LCOA (\$ > 0\$):** Deep decarbonization interventions (e.g. electric boilers, carbon capture) that require net capital subsidy.

6. Sector Inadmissibility: The Blocked & Capped Rules

 Toddler Version:

You shouldn't give a baby a giant heavy hammer, right? Same for factories! A computer shouldn't tell a medicine factory to use dirty recycled plastic for medicine bottles, or tell a tile factory to burn messy wood in a shiny tile kiln. Prangara has strict "safety locks" so it never gives silly recommendations!

Found in `backend/engine/macc.py` . A recommendation engine loses all executive credibility if it makes technically valid but practically illegal or dangerous suggestions:

Sector	Intervention	Rule Type	Engineering Justification
<code>pharma_formulation</code>	Recycled PET Substitution (<code>RPET_SUB</code>)	BLOCKED	Primary pharmaceutical packaging in direct product contact must meet US FDA / GMP standards. Recycled resin is inadmissible without full change-control stability testing.
<code>ceramics_tiles</code>	Biomass Boiler Switch (<code>BIOMASS_BOILER_SWITCH</code>)	BLOCKED	Tunnel kiln firing requires stable high-temperature flame and zero ash carryover onto glazed tile surfaces. Briquette firing ruins tile finishes.
<code>food_processing</code>	Recycled Resin (<code>RPET_SUB</code>)	CAPPED AT 35%	FSSAI food contact regulations restrict non-certified secondary resins. Substitution is modeled strictly on the non-contact secondary transit packaging.
<code>auto_components</code>	Secondary Steel Scrap (<code>STEEL_SCRAP_SUB</code>)	CAPPED AT 45%	Safety-critical automotive fatigue components carry strict OEM tensile standards that secondary melt steel cannot satisfy without metallurgical requalification.

7. Export Exposure Math: EU CBAM & SEBI BRSR

Located in `backend/engine/assess.py` (function `_compliance`). How does the backend predict cross-border export penalties?

```
$$\text{Embedded Operational Emissions } (tCO_{2e}) = \text{Scope 1} + \text{Scope 2}$$
$$\text{Exported Embedded Carbon } (tCO_{2e}) = (\text{Scope 1} + \text{Scope 2}) \times \text{EU Export Share } (\%)$$
$$\text{Indicative CBAM Annual Exposure } (\text{₹}) = \text{Exported Embedded Carbon} \times \text{Reference Carbon Price } (\text{₹7,000/tCO}_2\text{e})$$
```

Why Upstream Scope 3 is Excluded: CBAM rules require certified primary supplier data for precursor materials. Guessing upstream supply chain emissions would be presenting an estimate as a verified liability, which would fail customs scrutiny.

8. Backend Service Directory & File Map

File Path	Primary Responsibility	Key Algorithms & Invariants
backend/engine/assess.py	Master Orchestrator	Executes footprint $\$ \rightarrow \$$ leaks $\$ \rightarrow \$$ MACC recommendations $\$ \rightarrow \$$ Sankey flows $\$ \rightarrow \$$ CBAM in one pass.
backend/engine/footprint.py	GHG Accounting	Decomposes footprint by stream; calculates gate-to-gate intensity vs cradle-to-gate; isolates biogenic CO ₂ .
backend/engine/leaks.py	Diagnostic Rules	3-tier anomaly detector relative to BEE MSME sector benchmarks (p25, p50, p75 percentiles).
backend/engine/macc.py	Financial Economics	Annualized Capex (CRF), Levelized Cost of Abatement (LCOA), Simple Payback, sector blockage filters.
backend/engine/factors.py	Factor Registry	Stores official CEA v22 regional grid emission values, fuel coefficients, and uncertainty bands (low/base/high).
backend/app/services/report.py	Vector Report Engine	Generates inline vector SVG MACC charts and compiles auditable PDF working papers via headless Chromium.
backend/app/services/logistics_service.py	Freight Consolidation	Multi-factory milk-run pooling along industrial corridors; calculates modal shift carbon reductions.
backend/app/services/ollama_service.py	Private Vision Ingestion	Connects to local multimodal LLM on port 11434 to parse utility bills and machine nameplate photos.

Summary Invariant:

Every single number produced by Prangara can be audited by a chartered accountant or industrial inspector using a standard hand calculator and the published government Gazette. The AI acts only as eyes and ears; the brain is 100% auditable deterministic mathematics.